Input system:
Dependent variables:x_2, x_1
Independent variables:t
Equations:

$$\frac{dx_1}{dt} = ax_1^2\text{inv}(N + x_1) - bx_1x_2$$

$$\frac{dx_2}{dt} = (-1)cx_2 + dx_1x_2$$

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Result:
Dependent variables:x_2, s_1, x_1
Independent variables:t
Additional variables:

$$s_1 = \text{inv}(N + x_1)$$

Equations:

$$\frac{dx_1}{dt} = ax_1^2s_1 + (-1)bx_1x_2$$

$$\frac{dx_2}{dt} = dx_1x_2 + (-1)cx_2$$

$$\frac{ds_1}{dt} = (-1)s_1^3ax_1^2 + s_1^2bx_1x_2$$